JG University of Management

Department of Data Analytics

BORDER PROCESSING SYSTEM

LOAD AND FLOW ANALYSIS (2023–2026)

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1. INTRODUCTION:

The intake and transition of pediatric populations within border systems is a high-stakes operational and public health challenge. Tasked with the clinical intake and discharge of children, agencies like CBP and HHS operate a complex "patient" pipeline. Efficient data monitoring is critical to prevent facility saturation and ensure that medical and humanitarian care standards are met without delay.

Project Scope & Objectives

This study utilizes longitudinal data analysis to evaluate the throughput of minors through the federal care system. By treating the process as a care-delivery funnel, the project aims to:

- **Model Population Flow:** Quantify the "Length of Stay" (LOS) across different processing stages to identify systemic bottlenecks.
- **Predictive Trends:** Correlate historical volume surges with resource availability to forecast capacity needs.
- **Operational Efficiency:** Detect inefficiencies in the transition between intake (CBP) and specialized care (HHS) to optimize resource allocation and mitigate health risks.

2. PROBLEM STATEMENT:

Despite the continuous collection of operational data, the system lacks a centralized analytical framework to monitor clinical and logistical throughput. Currently, there is no standardized method to track:

- **Total system load:** Real-time visibility into total population volume.
- **Balance between intake and discharge:** Imbalances that trigger facility saturation.
- **Periods of capacity stress:** Data-driven indicators of critical system stress.
- **Sustainability of care delivery:** The impact of high volume on medical care quality.

The absence of these metrics results in reactive decision-making, leading to avoidable overcrowding and increased risks to pediatric health outcomes.

3. OBJECTIVES:

Primary Objectives

- Quantify total system load across CBP and HHS
- Identify capacity stress and relief periods
- Analyze balance between inflow (apprehension) and outflow (discharge)

Secondary Objectives

- Support planning for shelters and staffing
- Improve decision-making through data insights
- Enable data-driven humanitarian response evaluation

4. DATASET DESCRIPTION:

The dataset contains daily records including:

- Date
- Children apprehended and placed in CBP custody (inflow)
- Children in CBP custody
- Children transferred to HHS
- Children in HHS care
- Children discharged from HHS Care (outflow)

The data spans from 2023 to 2026 and represents system activity over time.

5. METHODOLOGY (EDA)

The analysis was performed in Tableau using a structured Exploratory Data Analysis (EDA) approach to evaluate system performance and trends.

Data Preparation:

- Time-Series Standardization: Converted raw date values into a consistent format to enable accurate time-based analysis.
- Data Cleaning: Addressed missing values and inconsistencies to ensure data quality.
- Calculated Metrics: Developed key indicators to support analysis:
    - Total System Load: Combined population in CBP and HHS at a given time.
    - Net Flow: Difference between inflow (apprehended) and outflow (discharged).
    - Transfer Efficiency: Rate of movement from CBP to HHS.

Visual Analysis:

- Inflow vs Outflow (Dual Axis): Highlights the relationship between intake and discharge to identify imbalance.
- CBP vs HHS Distribution: Shows how children are distributed across both systems.
- Total System Load Over Time: Tracks overall system burden and identifies peak periods.
- Monthly Apprehension Trend: Analyzes patterns in intake over time.

Tools Used:

- Excel (Data Cleaning)
- SQL (if used)
- Tableau (Visualization & Dashboard)

6. KEY INSIGHTS & CLINICAL LOGISTICS:

1. Growth in System Load (2023–2024)

- A notable increase in system load was observed, primarily due to higher intake compared to discharge.

2. Imbalance Between Inflow and Outflow

- In several periods, inflow exceeded outflow, resulting in accumulation within the system.

3. Capacity Stress Periods

- Peaks in total load indicate periods of increased pressure on system resources.

4. Stabilization in 2025

- A decline in both intake and total load suggests improved balance or reduced inflow.

5. Transfer Efficiency Observations

- Transfer rates above 100% indicate backlog clearance, where previously accumulated cases were processed.

7. RECOMMENDATIONS:

- **Enhance Transfer Efficiency:** Improve coordination between CBP and HHS to reduce delays.
- **Maintain Flow Balance:** Align discharge capacity with intake levels to prevent accumulation.
- **Plan for Peak Periods:** Use historical trends to support proactive resource planning.
- **Implement Real-Time Monitoring:** Utilize dashboards for continuous tracking and better decision-making.

8. CONCLUSION:

This analysis underscores the critical role of structured data architecture in managing high-stakes humanitarian logistics. By converting fragmented operational data into actionable intelligence, this project provides a blueprint for identifying systemic bottlenecks and forecasting capacity needs. Ultimately, these insights empower stakeholders to transition from reactive crisis management to a proactive, evidence-based framework that ensures both operational efficiency and the delivery of humane care.